Walks, Paths and Cycles Part One

Text
A walk from a node S to a node T in a graph is a sequence of successive edges that starts at S and ends at T. A walk may visit the same node more than once. For example, here you see a walk that visits node C more than once. A path is a work in which the intermediate nodes are distinct. For example, the path from C to E to F does not visit the same node more than once. A cycle on the other hand, is a path that starts and ends at the same node. For instance, here we have a cycle between nodes B,C, and D.

Walks, Paths and Cycles Part Two

Text
In the visualization that you see, we're asking how many works of length-3 are there from node A to node C or from A to node D in this graph? The answer is given by the element sd of the matrix A to the kth power, where k is the walk length. So what you see in this visualization shows the cubic power of the adjacency matrix A. And that shows there are four walks actually of length-3 from A to C, the walks ABDC, ABAC, ACAC and ACDC. On the other hand, as you can see, the number of walks of length-3 from A to D is 0. There are no walks of length-3 that can take you from node A to node D.